lecture 1

First, you should understand the background and moral structure of the authors. So if you read a treatise, translation paper is easy. You call it a file of information.

Try to understand the background and the motivations. Why they study this problem or develop this technique. And you should understand the technical details.

And why their work is known or not known. Are their works very important? Or do they have any important impacts? So then you can ask yourself. Why these guys are able to generate such ideas? So why? You should think about the reasons.

So why these works are important or difficult. So this is about if you were a survivor, what comments can you give to these authors? If you were an author, can you do better? You should carefully think about these issues. Which will help you to get a deep understanding of a paper or area.

So how to choose a research topic? First of all, you should follow your survivor. Because your survivor pays money to you. Without money, you cannot do research.

This is obvious. However, you still have freedom. So what is a good area? An area which is important but not well-developed.

This is a very good area. Some areas are very important, but you are not able to do research. For example, high school mathematics.

It is very important, but there is no room for you to do research. Which is not a good area. So basically, you do the most important work which you are able to do.

You should try to maximize the importance of the topic. Something to make it complete. For example, your ability.

And your survivor's suggestions and so on. So how do we understand the importance of a topic? You should have a good understanding of the whole area. So why is this area important or not? Not all areas are equally important.

Some areas definitely are not important. So you should improve your ability. So this is my suggestion.

Think about three or more different topics at the same time. So you can cross over different ideas. So how to write a paper in English? So English is not our mother language.

But we have to write papers in English. You should study the writing styles of quality papers. I mean good papers, not rubbish papers.

A lot of papers are rubbish. You should read some quality papers. Carefully analyze the writing style.

So how to read a paper? Because you will be a writer. You will write these papers. So do it as a director watches films by others.

So be very careful. So you need to learn the structure of a paper. The main topic of this paper.

So can you use a few one to ten sentences to describe this paper? And the main points of each section. Every paper has maybe five sections. For every section, you should try to work out the main points.

For every paragraph, you should be able to work out the main points. Then you should think about, can you make these points clear? Much more clear than the author. Can you improve it? Can you optimize or improve the structure? Can you do better? So how can you write these paragraphs, sections, papers in your way? Much better than others.

So gradually, you will form your style. You should fix your writing style. Every paper, you have novel ideas.

Excellent research results. But your writing style should be fixed. Because English is our second language.

So try to have your bubble. And improve your style. Just fix it.

So this is important. This is my suggestion. So what is good writing? So we all use a newspaper, textbooks.

So not a piece of hard work. It will be read by others who cannot meet you face to face. So it's not easy to write a good paper.

So you should think about the structure first. So make your paper as imaginative as possible. Think about who will be your readers.

So try to make your paper attractive. And you should try to make your paper as accurate as possible. Every sentence should be very clear.

Have a unique meaning. So the reader can easily understand the poems you want to make. For example, you write one paragraph.

You want to make two poems. Even people can get different poems for your paragraph. Meaning something wrong.

All people can easily understand you try to make two poems. In this paragraph. So my way is to use bullet points.

I often write it this way. I try to make two poems. Point one, point two.

To help readers understand my poems. I usually use standard concepts. So every concept should be very clear.

To people researching your area. So try not to create new concepts. If it is not absolutely mainstream.

Use all the concepts. All the technologies to describe new ideas. So you should optimize the methodical applications.

For example, AX equals B. X as a null. A as a constant. Try not to use A for a null.

X as a constant. Use the following tradition. So sometimes, maybe it is very difficult for others to understand your ideas.

You should give some simple examples in your paper. Or in your thesis. To help others understand your poems.

This is my experience. So finally, try to be a good person. Be helpful and friendly to others.

Because we hope others can be friendly to us. We should do the same. So second most important.

Even more important than your research. Try to build your health habit. And you should self-motivate.

And work hard. Try to be a deep thinker. Because you want to do research.

You want to produce knowledge for the world. You are a guy who can produce ideas. So try to be the best.

Try to be better than your survival. This is your first target. OK.

I know everyone in this room understands Chinese. OK. These two sentences are for you.

OK. OK. This is the first part.

OK. I still have some time. Now I will use some of my books.

So we will have a five-minute break. Or not. I swear.

Because we are very tired. Do you need a break or not? In case you ask me, so I will need a break. So let's have a five-minute break.

You can talk to each other. Tell your neighbors to make friends. OK.

So I also have a five-minute break. I'm tired. You add me on WeChat.

Are you in that group? CS. Yes. Computer.

OK. Non-official group. Don't you know this group? The group with 229 people.

No, this group. I'm in it. That's great.

I'm not in it. Let me add you. You can add me.

You know what, we haven't started yet. We have 8 I'm in it. singing I am sorry, I didn't catch your voice you were keep on spinning What direction are you heading to? I'm going to be a naturalist.

Which director? Song Linqi. I'm going to be an AI professor. I'm going to be a computer scientist.

And I'm going to be in charge of the project you want me to be in charge of. Isn't that the direction you were in last year when you won the Nobel Prize? Yeah, yeah, yeah. Almost.

Yeah. I'm going to be a computer scientist. I'm going to be a computer scientist.

So which project are you working on now? I was working on a legal model before. I'm doing something else right now. But I haven't started thinking about it yet.

Okay. We have an informal web chat. Who? For you? Okay.

So this is the model. CTU says their office and the PhD students. Okay.

So how long do you want to join? Okay. Okay. Okay.

Okay. Okay. Okay.

Okay. Okay. Okay.

Okay. Okay. Okay.

Okay. Okay. Okay.

Okay. Okay. Thank you.

Why is it so hot? Why is it so hot? My bag can't go out. But it's hot. My bag is more comfortable.

His bag is more comfortable. Do you want to go in? He is singing. He is singing.

Hi, I heard you said you wanted to share your skirt with me. Hi, I heard you said you wanted to share your skirt with me. My god, I need a half-hour.

So actually, this was my talk last week for Huawei. So, I think you should understand your area. The research topics, projects in your group.

You should also know something about articles in our department. This is one of the topics in this talk. So, my major research areas are optimization.

I believe everyone will use optimization. Which is one of the most important mathematical techniques, particularly for AI. Do you know what is optimization? This word.

What is optimization? Try to make the word better, okay? Over the last 30 years, I've been working in this area for a very long time. So, one of my research topics is a framework called IMO-EAD for multi-optimization. So, I will give you maybe 10 or 15 minutes.

I hope it's useful for you. For you to understand what I'm doing. Over the last 5 years, one of the major research topics in my group is automatic algorithm design.

We try to do it by AI to design algorithms. We propose a method called UIA. So, you know where we are.

This is my office. You know this building. This is my room.

Okay. So, I will talk about three things. What is IMO-EAD? And some new developments on this framework or this algorithm.

And the last one, UIH. How we use a large lambda model to design algorithms automatically. So, this is my research problem.

Multi-Optimization Problem. Multi-Optimization Problem. We try to minimize or maximize a few functions.

Each function is called an objective function. So, every function is a target. So, I have maybe two, maybe three.

So, we have a few objectives. Objectives. So, X is a decision variable.

X takes value from D. We call it the source space. So, X can be discrete, 0, 1, or integer. Or continuous variable.

Or mixed variable. Actually, X can be in any data structure. X even can be a tree, a neural network.

Okay, in our research, X can be an algorithm. Okay, can be a computer program. So, X can be in any data structure.

If you can put it in your computer memory, it's okay. So, first of all, why do we care about multi-objective optimization? Why do you choose this topic? You always need to ask why we need to study this topic. Why? Why is this problem important? So, I have three basic reasons.

So, by nature, many real-life problems and multiples. For example, if you design a neural network, you always try to make the neural network as simple as possible. Okay? And its performance should be as good as possible.

So, there are two different objectives. How do we solve it? In university, most of you took a so-called mathematical modeling course. Have you taken it? Mathematical modeling.

So, if you do mathematical modeling, if you have two objectives, what do you do? Tell me what you do. If you have two different objectives. So, you should try to combine these two objectives.

So, if you have two objectives, what should you do? You should combine them into one. Then you can solve it. But sometimes, because you have not enough information, you don't know how to combine them together.

So, in a sense, modeling has not yet finished for this problem. We don't know how to combine them together. In some cases, even you know how to combine them into one single objective, but you want to know more about tradeoff relationship among F1 and F2.

So, in this case, you must use this technique. You must use multiple objective optimization techniques. So, first of all, you need to know how to compare the qualities of two solutions.

There are many ways. The most commonly used way is to use dominance. So, I'll take two objective examples.

F1 and F2. We try to minimize F1 and F2. So, this side is x-space.

This side is F1 and F2-space. So, F1 and F2 are objectives. So, given two solutions, y dominates x, means y is better than x, if and only if, this is the definition.

y is no worse than x in any objective case. And y is better than x in at least one objective case. So, this is the definition.

So, in this example, these solutions, solutions A, B, C, these are objectives. So, let's compare A and B. Which one is better? We try to do minimization. The smaller the better.

B is better or A is better? B. So, B is better than A. So, how about B and C? Think about it. How do you compare B and C? B is better or C is better? Actually, according to this definition, we are not able to compare B and C. So, in terms of F1, in terms of F1, B is better. In terms of F2, C is better.

But according to this definition, we are not able to compare B and C. So, in mathematics, this kind of relationship, called partial ordering, is not complete ordering. So, sometimes, for two items, you are not able to compare their qualities. They are partial ordering.

This makes multi-objective organization more difficult than single-objective organization. So, using this definition, we can define optimality. A question of solution.

Again, I'll use an example. Suppose we have two pieces F1 and F2. So, a solution, X, is a partial optimum.

If and only if no other solution dominates it. Means you have a solution. Your solution is a partial optimum.

I have a solution. If we compare my solution and your solution, there are two possibilities. Number one, your solution is better.

Number two, our two solutions cannot be compared. This is a partial optimum. So, of course, a rational decision maker does not like no partial optimum solution.

Because if a solution is not partial optimum, we can improve one objective without making the other objective worse. You can improve F1 without making F2 worse. So this is why people like partial optimum solutions.

So if you have one objective, in most cases, you only have one single globally optimal solution. Why? Why if you have one objective, in most cases, you only have one single globally optimal solution. Or, very few globally optimal solutions.

In most cases, why? So think about it, okay? In your own time. So this is my equation. There are some mathematical reasons.

But, in multi-objective optimization, we should have a lot of different optimal solutions. So I put all the solutions together. I call it the Pareto set.

So Pareto set is a set of optimal solutions in decision space, in F space. Okay? So there's the PS. So PS will have an image in F space.

So PS has an image of PS in the objective space. So PS has very good meaning in geometry. So this area are feasible solutions.

All solutions. Actually, what is PS? We do minimization. PS is lower left boundary.

Okay? This boundary. This red curve. Okay.

So multi-objective optimization is not new. It has over 100 years history. Okay? Very old areas.

So why do I still study it? Okay, you can also think about it. So over the last 100 years, or at least 80 years, there are many different methods have been proposed for solving this problem. Okay.

I think there are more than 1,000 different algorithms. But in any subject, in any area, okay, we cannot induce this 1,000 algorithms to use one by one. Okay, you have no time.

We should do classification. Okay? This applies to all the areas. In all the areas, we only have very few principles or basic strategies.

Why I'm calling it basic strategies? It cannot have more than 10. Okay? Maybe 3, maybe 4, maybe 5. Okay? If more than 10 means some strategies are not basic, this is true for all the computer science areas. You can only have very few basic techniques, basic principles.

In this area, though we have more than 1,000 different methods, this method can be divided into 3 categories. Okay? The first one, very easy to understand. I don't know how to solve multi-objective online problem.

I just transform this problem into a single game problem. This is a commonly used trick. For a new problem, I don't know how to solve it.

I just transform this new problem into a problem I can use. So, I can I have two objectives I just combine them together. I got one single objective, fx.

Then, I can use a single objective over there, store it. So, in a sense, I pick up a point from a PS and give this point to its maker. Okay? This is one of the commonly used strategies in this area.

It's very simple. Okay? The second one I use over there to produce a number of patterns of solutions maybe 100, maybe 20 and some other information to cover to approximate the whole platform. And then I pass all these solutions and some useful information to its maker.

Its maker may be a group of people, may be a computer programmer. Okay? Maybe a machine. So, its maker receives these solutions then he can have a good understanding of her problem.

Then, he can choose a solution she likes most. Okay? This is the second methodology. The last one is interactive methodology.

We first produce a number of solutions. These solutions can be optimal or not optimal. And I give these solutions to its maker.

Its maker because he or she has more information about a problem. I can tell all of them her preference. For example, she likes this area.

Then all of them search this area. Get more solutions. So after a number of iterations, we hope the decision maker will get the solution she likes most.

Still there are a lot of problems. So, can you guarantee in this way, finally, the decision maker will get the solution she likes most? Basically, we have three strategies over the last 100 years. So, over the last 30 or 40 years, all of them become very popular for solving multi-objective operations.

So, for me, most of EAs actually are population-based iterative search methods. So, it works in this way. Suppose you want to solve a problem.

Your problem may be single objective, may be multi-objective problem. Any problem. You do iteration.

At each iteration, you keep a number of solutions, a number of candidate solutions. For example, one solution. So, this one solution requires population.

Okay? Each solution is a candidate solution. So, select to your problem. Your problem may be an open-ended problem, may be not.

Okay, just a problem. And then, next step, you do selection. From this one solution, you select 50 good solutions.

Okay? So, these 50 solutions will be paired for next generation. Then, next step, you do crossover of parent solutions. So, you mix two parents, or three parents, or even many parents together to generate one new solution.

This is called crossover. You can also do mutation. You modify a single solution to get a new solution.

Such as local search, or gradient search, given a solution. You modify it a little bit to get a new solution. So, this is cross mutation.

So, now you get one new solution. This one new solution will be solution for next generation. Then, you can do iteration for one new generation for April.

So, finally, we hope we can get a good solution to our problem. This is the basic framework. So, actually, some modern years also contain models and some other memory.

So, we can keep some models to summarize good features or bad features of solutions. So, if a solution has this feature, then, it's very likely it will be a good solution. So, you can organize this feature.

Remember these features using a model or using some other memory technique, for example, tabu memory. So, what is the advantage of this approach? It's easy to use. So, a solution can be in any structure, even a language model, okay, as did in some companies.

And suitable for black or green box problems. Even if you know very little about your problem, okay, if your mathematics is not good, you can use this technique. So, this technique becomes very popular for solving multiplicative or random problems.

So, we have an area called the YIMO area. So, a YIMO method is to produce a number of solutions to approximate the problem or provide some other useful information to data makers, which is very popular in many areas. A lot of companies use this technique, including many big names.

It becomes a mainstream methodology in most criteria in the making area. Okay. This is a mainstream technique for this reason, because it's important.

So, a lot of people are doing research in this area. So, it becomes the hottest research area in the world. So, is the main reason why we doing Okay.

the main reason why we are research. Okay. OK, again, we can classify these different algorithms into three categories.

OK, this is science. You have one sort of different method used to do classification. So in this area, people classify all the algorithms in three categories.

OK, this was done in email conference, in the flagship conference in this area, not by me, OK? So three categories, OK? The last one was proposed by me, OK? It's become an area, become a research area. It was proposed by me a long time ago, OK? So now I will use this area to you. Hope it's useful for you to do your research.

It's called the Dignation-Based Algorithm. So it's one of the most researched and used frameworks in this area over the last 10 years. OK, what's the ideas behind the MOAD, Multi-Objective Elimination Algorithm, based on Dignation? So behind this technique, only two components.

One is Dignation. The other is a collaboration. So collaboration is super important, OK? OK, so what's my task? My task actually is to produce a number of points to approximate the platform, this black curve, OK? This is my task.

Wow. I decompose this task into n sub-problems, into 20 different problems, OK? I have one task. I divide this task into 100 sub-problems.

Each sub-problem can be a single objective or multi-objective. So this is the first component. So this strategy is widely used in computer science area, OK? Dignation.

The second technique is called collaboration. It's for swarming projects. I have 100 problems, then I use 100 agents for solving these 100 problems.

Each agent is an object for a different problem, OK? So because these 100 problems are related to each other's work, why can I claim these 100 problems are related to each other? Because they are from the same problem, OK? So definitely, they are related to each other. So I can use the, so this 100 agents can help each other because their problems are the same, OK? So let's get an idea. So I use this diagram to explain my idea.

Maybe it's helpful for you. You can use the same technique. So given a problem, in my case, it's a multi-objective, multi-agent problem.

I decompose it into 100 single objective problems. Then I use some conversion techniques to solve these 100 problems. Then I have 100 different solutions, OK? These 100 different solutions can be used to approximate the problem.

So that's my idea. How to implement this idea? How to transform this idea into an algorithm? You need to solve the following three issues. The first one, how you decompose it into 100 problems, OK? How we define relativity among these 100 problems? We use neighborhood structure.

You can also use these other concepts, OK? I use neighborhood to describe relativity among different problems. Issue one. Issue two.

For each subproblem, what method will be used for solving it? So if we use a search method, if we use iterative search, actually there are two things you need to consider. One is the memory. At each iteration, what kind of information should be recorded? And how can you use information to produce a new solution? So two issues, two sub-issues.

The last one is calibration. How different agents work together, how to charge them? OK? If you solve these three issues, you will design a MOEAD object. This is why we have a lot of different MOEAD objects, OK? So in the following, I will show you an example, OK? Are you tired? Like me.

Are you very tired? Do we need a five-minute break? OK, let's save you five minutes, OK? For five minutes, OK? Which problem has a width? We can define neighborhoods based on widths. So we have a problem with five, has five neighbors. Why these five problems are neighbors of problem five? Because these five problems are very close width-vectors to problem five.

So these five problems are similar, OK, in terms of a landscape, or optimization, or optimal solutions. Of course, there are many ways for defining neighborhood structure. Then we can have a very simple already over that.

So I have one of the problems. I use one of the agents, OK? We define neighborhood structure among all the problems and all the neighbors, all the agents. So each agent use iterative search to solve its problem.

To make the problem, to make all of them as simple as possible, each agent only record one solution during the search. For example, agent one only record x1 for problem one. x1 is the best solution found so far for problem one.

Agent two only records x2. x2 is the best solution found so far for problem two. Everyone will do iterative search.

Now the problem is at each iteration of iteration, how each agent generate a new solution. Suppose I use genetic algorithm, OK? Because each agent need more than one solution for doing crossover. Suppose I'm agent five.

My problem is problem five. I want to solve problem five. I only have one solution x5.

So I'm not able to do crossover. But I have neighbors, OK? I can borrow some solutions from my neighbors. Problem four, problem eight.

So my neighbor's problems are different from my problems. But our problems are similar. So their solutions also useful for me, OK? This is the society.

So everyone has his own target. Different person, different people have different targets. But our targets are the same.

This is what we can learn from each other, OK? So agent five collects some solutions. He can use these solutions to generate a new solution. The next step, if this solution is better than his current solution x5 for problem five, then he can use this new solution to replace his old solution, OK? Update x5.

Because his friends have helped him. He need to help his friends. He can pass this new solution on to his neighbors, OK? Each neighbor can decide if this new solution can be used to replace their old solutions.

So this is basically an overdub. It's very easy to implement this overdub, OK? So I will not go into details. You can add a lot of techniques to this framework.

So this is the impact of this method. It has become one of the two most used frameworks in this area. It has been included in some textbooks.

OK, it's citation is still going up, OK? Every year, the citation is about 1,000, still very hot. And so why is it so hot? Because multi-objective automation becomes more important than before, OK? So it's not a new technique. So this is a Google website developed by two guys in France.

So there are a lot of survey papers on this area. More than 10 in English. More than 10 in Chinese, OK? So there's some recent survey papers.

So for example, this one is about the application of MLAD in machine learning. And also a lot of applications in industry, OK? So now I try to use a, OK, this is some new development. So I don't have time for this part.

For example, you can use it for larger model alignment, OK? Or drug design, and so on. There's some new concept in this area. So I try to use five or eight minutes to use my recent work, erosion of a problem, circle-happy problem.

So given a square, can you put 26 N circles in this square? Such that the sum of the radius is maximized, OK? This is a classical problem. So we can use our framework, our methodology to solve this problem, to produce a new algorithm to get the best result, OK? This is our result, OK, which is the best in the world, OK? This is ours. And we can write the paper, not by student, but machine, OK? For another model.

So I guess it's eight or 12 pages. Looks like a paper, OK? So this is the framework, OK? This is the platform. It has been used for solving 100 different tasks.

You can try, OK? It's open source. So here's some papers, OK? So this is my major work. One is multi-objective optimization.

The other is how to use AR to design algorithms to replace myself. OK, so that's all, OK? So if you have any problems about optimization, you are welcome to go to my office. Do you know where my office is? In this building, OK? Level seven.

Between lift six and seven. OK, so see you around.